dence of the challenges," says Godse. According to Deloitte, 68 per cent of Indian organisations will place greater emphasis on security and compliance controls when investing in AI, while 94 per cent intend to increase their investment in the field over the next year.

According to Jagannarayan Padmanabhan, senior director and global head, consulting, at Crisil Intelligence, compared with other sectors, AI adoption in infrastructure remains low. "The regulatory framework required to determine how and where AI should be used, and where its use needs to be monitored, is still to come. I would assume that, at least initially, this may be more self-driven rather than imposed at the policy or government level," he says.

Months before the Hugging Face break-in, the Modi government's India AI Governance Guidelines, drafted in November 2025, identified establishing key governance institutions and developing India-specific risk frameworks as immediate priorities. Its medium-term goals included publishing common standards, amending laws and regulations, operationalising AI incident-reporting systems, piloting regulatory sandboxes and expanding the integration of India's digital public infrastructure (DPI) with AI.

Manufacturing Risks

Manufacturing was among the three most targeted sectors in India by threat volume this year, with phishing attachments and compromised design tools common entry points, and ransomware the highest-impact category despite its lower volume. The concern is compounded as Indian manufacturers wire AI into decision-making, predictive maintenance, quality control and robotics: the models themselves become targets, since 'poisoned' data can quietly degrade decisions. "The principal security concern is not any single threat but the combination of cyberattacks, supply-chain compromise, data leakage, intelligence manipulation and loss of operational control," says Bikash Barai, co-founder & CEO of cybersecurity startup FireCompass. "A compromised AI system could expose sensitive information or influence decision-making, while uncertainty over how an AI system is functioning, where its data is going, or whether it has been manipulated could become a serious national security risk."

In cybersecurity parlance, 'attack surface' is a key term: the fewer doors a network has, the better. "The integration of IT systems with operational technology is making manufacturing more efficient, but it is also creating more points of entry for cyberattacks.



> "Until now, we had a small number of actions to control. With AI, we must be **mindful of the scale, complexity and interdependence** of the challenges"
>
> **VINAYAK GODSE,** *CEO, DSCI*

This is happening faster than companies, especially MSMEs, can strengthen their cybersecurity systems," says Gerald Beuchelt, chief information security officer (CISO) at the Switzerland-headquartered cyber protection firm Acronis. Acronis's threat research and India-focused data, Beuchelt adds, indicate AI is compressing attackers' timelines. Its Threat Research Unit recently documented a China-linked group building and deploying malware referencing a major world event within hours of news of it breaking. "We expect similar speed to be used against Indian targets, as AI now lets attackers craft convincing, well-timed lures fast," he says.

The Opportunities

But here's where AI comes in again, through red-teaming—the industry term for testing cybersecurity readiness by simulating attacks. This year, CERT-In, the national nodal agency in India responsible for dealing with cybersecurity threats, hacking and emergency computer security incidents, conducted 10 AI-threat cybersecurity exercises between June and July, involving 1,470 participants from 345 government and private organisations. AI, it says, can help reduce defensive response times by automating the discovery of malicious domains, phishing attempts and vulnerabilities. Its frontier-AI advisory calls for external AI red-teaming, micro-segmentation, robust identity controls and multi-incident exercises. As Rajesh Nambiar of Nasscom notes, this is essentially the next evolution of cybersecurity as a service—and that is where the market is headed. "The opportunity for India therefore goes beyond defending domestic systems. It is also to build the next generation of cybersecurity services for a world adopting AI at scale," he says.

Scale is possibly the key word. "Deployment of AI is actually hard, and India is good at building digital systems," says technology lawyer Rahul Matthan, who was part of the RBI committee on the Framework for Responsible and Ethical Enablement of Artificial Intelligence (FREE-AI). AI, he adds, is here to stay. "In that case, what does India do? We should learn to optimise it for our goals and objectives in the safest possible way." The FREE-AI report, in fact, calls for a framework built around two themes: innovation enablement and risk mitigation. In a world fearful of AI agents going rogue, the other side of the coin is the risk of missing the bus on a transformational technology—one that could itself provide the guardrails against such superintelligence. India should see this as another Y2K opportunity and establish areas of AI dominance of its own. ■

with ***Pradip R. Sagar, Avishek G. Dastidar & Sonal Khetarpal***